Sahaj Bhadja

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Summary

Software engineer with 2+ years building SQL-backed applications, healthcare workflows, REST services, and internal tools. Experienced in Java, Python, JavaScript, structured programming, relational databases, testing, debugging, secure access, documentation, Git, Agile, and production support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and internal dashboards for 500+ users, translating defined requirements into maintainable workflows while reducing production latency from 1.2 seconds to 350 milliseconds.
- Orchestrated 5,000+ monthly jobs through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving API observability, operational resilience, recovery, maintainability, and systematic production issue diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask REST services handling 10,000+ daily requests, implementing defined workflows, authentication, error handling, and reusable components while resolving production defects and supporting 150+ users.
- Created multithreaded processing for 1,000+ files with validation, AWS RDS persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining responsive behavior under concurrent production load.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services and SQL integrations using object-oriented patterns, filtering, pagination, caching, AWS Glue, and S3, improving production response times 30% and data-processing efficiency 68% across applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, documenting test evidence while accelerating defect correction, release verification, and reliable deployments across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js features across eight microservices with authentication, role-based access, documented business workflows, and real-time updates, securely processing 2 million+ Azure SQL application records.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving service reliability, deployment consistency, and operational visibility.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built React Native clients for iOS and Android with Firebase persistence, real-time state synchronization, image storage, push notifications, and secure sessions across a mobile platform handling 50,000+ records.
- Developed Flask REST APIs for bookings, pricing, and payments, delivering sub-200-millisecond responses and reliable webhook-driven external integrations that improved confirmation speed 40% across customer-facing mobile application workflows.

Skills

Languages & Fundamentals: Java, Python, JavaScript, TypeScript, SQL, C++, structured programming, object-oriented programming, variables, loops, procedures, data structures, algorithms
Applications & Security: React, Node.js, Flask, FastAPI, REST APIs, web services, microservices, authentication, authorization, role-based access, secure sessions, consent-based access
Databases & Cloud: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, Firebase, AWS RDS, AWS, Azure, relational databases, schema design, query optimization, caching
Quality & Delivery: Git, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, functional testing, validation, debugging, error handling, defect correction, code review, test documentation, production support, Agile, Scrum

Projects

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases with schema-aware query generation, structured validation, optimized execution, metadata exploration, and Docker-backed deployment across modular Java and TypeScript services.

CPU-Scheduling Simulator | Live | *C++, React, JavaScript, Algorithms, Testing*
- Built a C++ simulator implementing six scheduling algorithms with deterministic workloads, edge-case validation, Gantt visualization, and performance testing, reducing average turnaround time 25% across 1,000+ process configurations.

HMIS | Live | *TypeScript, React, REST APIs, Appwrite, Sentry, Twilio*
- Built responsive healthcare workflows for appointments, records, and consent-based access using React, REST APIs, authentication, Twilio notifications, and Sentry monitoring, supporting 100+ daily interactions while reducing administrative workload 30%.